

Heat decarbonisation module: specification

Asset register, gas data and indicative heat pump sizing against electrical headroom

In one paragraph

Add three things to the Data Analyser, in order. (1) A short asset register for existing CHP, solar PV and heat pumps, so headroom and data checks reflect what's on site. (2) Gas meter data, half-hourly or daily, analysed like electricity. (3) Indicative heat pump sizing: heat demand from gas, adjusted for weather, turned into a half-hourly electrical load and tested against the site's supply capacity. The headline output is a coverage curve: heat pump size against the share of annual heat it covers, marked fits or doesn't fit the existing supply. It stays a screening tool, offline, with every assumption visible and editable.

1. Purpose and audience

- **Question it answers:** roughly how big a heat pump does this site need, how much of its heat could it cover, and can the existing electricity supply take it?
- **Used by:** engineers and project development at feasibility and bid stage, including funding applications. Sales use the outputs in proposals.
- **Not for:** detailed design, equipment selection or DNO applications. Those still need a heat loss survey and a design.

2. Scope by phase

Phase	Delivers	Rough size
1. Asset register	Per site: existing CHP, solar PV and heat pumps. Headroom with CHP off. PV-aware data checks (export no longer flagged as a fault). Install dates shown on the heatmap and timeline to explain step changes.	Small, 2 to 3 days
2. Gas data	Load gas files (half-hourly or daily, kWh or volume). Quality checks, heatmap, profiles and gas findings: out-of-hours heating, summer base (hot water and catering), heating in warm weather. Weather link via degree days.	Medium, about 1 week
3. Heat pump sizing	Weather-adjusted heat demand, coverage curve, shaped heat pump electrical load added to the electricity profile, fit against headroom, running cost and carbon change.	Larger, about 2 weeks including validation

Sizes are build estimates only and assume test data is available. Phase 3 depends on 1 and 2; phase 1 is worth doing on its own.

3. Inputs

Asset register (per site)

Field	Notes
Type	CHP, solar PV, heat pump (air or ground source), other generation.
Size	kWe for CHP and PV (kWp); heat pump thermal kW and, if known, electrical input kW.
Install date	Optional. Used to mark step changes and to split before and after.

Field	Notes
Behind the meter?	Yes if its output or demand shows in the meter file being analysed. Only behind-the-meter assets change the electricity analysis.
Notes	Free text: run strategy, condition, separate metering.

Gas data

- **Half-hourly or daily.** Many public sector sites only have daily or monthly reads; daily must be supported. Monthly data is out of scope (too coarse to size from).
- **Units.** kWh, m³ or hundreds of cubic feet. Volume converts as $\text{kWh} = \text{m}^3 \times 1.02264 \times \text{CV} \div 3.6$, with the calorific value (CV, MJ/m³) from the bill. Default CV 39.5, editable. One hundred cubic feet = 2.83 m³.
- **Pairing.** A gas file is linked to an electricity site by the user. Different date ranges are allowed; see method step 6.
- **Non-heating gas.** Catering or process gas on the same meter, as an editable percentage or estimated from the summer base (method step 2).

Weather

- Heating degree days (base 15.5°C, the UK convention) for the site's region and the gas data's dates.
- **Offline constraint:** the app can't fetch these live. Either bundle a regional table in the app, or let the user paste a column of daily degree days. See open question Q5 on data source and licensing.

4. Method

- 1 **Heat demand.** Heat (kWh) = gas (kWh) × boiler seasonal efficiency. Default 80%, editable.
- 2 **Split heating from hot water and other use.** Summer weekday gas use (June to August) is treated as hot water, catering and process, assumed flat all year. Heating = heat demand minus that base. The user decides whether hot water is in the heat pump's scope.
- 3 **Weather adjustment.** Regress daily heating demand against daily degree days, occupied days only: heating = $a + b \times \text{HDD}$. Check the fit (R^2) and show it; a poor fit (below 0.7) is flagged, as it usually means controls problems or data issues. Design-day heating = $a + b \times (15.5 - \text{design temperature})$. Default design temperature −3°C, editable by region.
- 4 **Size from daily demand, not the gas peak.** Gas half-hourly peaks reflect boiler oversizing and morning boost, not the building's need. Design-day heat is spread over the heat pump's run hours (default 20 hours a day, assuming buffer storage). Heat pump thermal size for full coverage = design-day heat ÷ run hours. The gas half-hourly peak is shown alongside for comparison.
- 5 **Coverage curve.** For candidate sizes from 10% to 120% of the full-coverage size, simulate each day of the year: heat delivered = the lesser of that day's demand and size × run hours. Coverage = delivered ÷ annual heat. The boilers make up the rest.
- 6 **Electrical load.** Heat pump electricity = heat delivered ÷ COP. COP varies with outdoor temperature (daily mean taken as $15.5 - \text{HDD}$ when $\text{HDD} > 0$) and flow temperature, from an editable curve with a default per flow temperature band (Q2). The heat is spread across the run window to give a half-hourly load. Where gas and electricity cover the same dates, add date by date. Where they don't, match by month and day type and say so on the output.
- 7 **Fit against supply.** Add the heat pump load to the electricity profile and run the existing headroom check, with behind-the-meter CHP assumed off (import + CHP kWe) and PV assumed absent at the winter peak. Report fit or no fit, half hours over, worst clash and peak kVA with the heat pump.
- 8 **Cost and carbon.** Gas saved, extra electricity, net running cost at the entered gas and electricity rates, and carbon change using the current DESNZ conversion factors (year shown, editable).

5. Outputs

- **Coverage curve:** heat pump size (kW thermal) against share of annual heat covered, each point marked fits or doesn't fit within usable capacity. Lines at 100%, 90% and 80% coverage.
- **Options table:** for 100%, 90% and 80% coverage and the largest size that fits: thermal kW, electrical kW at design day, peak kVA with the heat pump, fits yes or no, gas saved, extra electricity, net cost change, carbon change.
- **Weather chart:** daily heating against degree days with the fitted line, R^2 and the design-day point.
- **Profile chart:** the design-week electricity profile with and without the heat pump, against usable capacity.
- **Assumptions panel** listing every value used, printed on every output.
- All of it in the site PDF, the Excel export and customer editions, where heat pump settings are read-only like the rest.

Caveat wording (on screen and on every output)

"Indicative heat pump sizing based on metered gas use and the assumptions shown. This is not a design. Sizing, coverage and running costs are subject to a heat loss survey, emitter and flow temperature assessment, hot water strategy and confirmation of supply capacity with the DNO."

6. Proposed defaults for engineering to confirm

Assumption	Proposed default	Why, and what to check
Boiler seasonal efficiency	80%	Between older non-condensing and condensing plant. Should the default depend on boiler age?
Gas calorific value	39.5 MJ/m ³	Typical UK value; the bill gives the actual figure.
Degree-day base	15.5°C	UK convention.
Design outdoor temperature	-3°C	Placeholder. Engineering to give regional values and the source.
Heat pump run hours, design day	20 hours	Assumes buffer storage. Should it be lower without storage?
COP curve	By flow temperature band	Engineering to supply curves for, say, 45°C, 55°C and 65°C flow, air and ground source.
Weather fit threshold	R^2 0.7	Below this, flag the result as unreliable.
Supply safety margin	10%	As now in headroom.

7. Validation before release

- Two or three real gas files (half-hourly and daily) from different building types, with bills for the calorific value.
- At least one site with a heat pump already installed and metered: compare predicted heat pump electricity with what was actually used.
- One site with CHP: confirm the CHP-off headroom against what the site team knows about trips.
- Engineering sign-off of the method and defaults in section 6, recorded with a version number.

8. Open questions

#	Question	For
Q1	Is hot water in scope for the heat pump by default, or heating only?	Engineering

#	Question	For
Q2	COP curves and flow temperature bands to use. Do we have manufacturer data we can embed?	Engineering
Q3	Should hybrid (heat pump plus retained boilers) be the default framing, or full replacement?	Engineering and sales
Q4	Regional design temperatures: which source, and at what granularity?	Engineering
Q5	Degree-day data: which source, and can we bundle it in a tool we hand to clients (licence terms)?	Engineering and legal
Q6	Should funding scheme outputs (for example carbon cost per tonne) be included? Which current schemes and rules?	Project development
Q7	Who signs off the method, and who owns the defaults when they change?	Martyn

What we need to start

Answers to Q1, Q2 and Q7, plus the test data in section 7. Phase 1 (asset register) can start now without any of it.